

Labeling the 45k dataset — MedGemma vs Mistral

We ran our EEG-report classifier over the whole processed_reports set (45,545 reports) and compared it against the Mistral-7B labels that already ship with the project. Both models label the same reports, so we can see where they disagree; and on the ~2,500 reports that also have a human label, we can check which model comes closer to the human.

Our model is MedGemma-27B with the v5 prompt and the consistency grammar (v5g), run at the Q4 quantization. We used Q4 because it holds up best on reports unlike the ones we tuned on, and this set is exactly that case: hospitals and physicians the model has not seen before.

What the labels are. Each report gets five findings — overall abnormality, focal and generalized epileptiform activity, focal and generalized non-epileptiform (slowing) — and each is scored 1 to 4. Everywhere in this report, **“present” means the finding was scored 3 or 4** (the model thinks it is there); a score of 1 or 2 means absent. All the percentages below are built on that present/absent call.

Accuracy against the human annotator

2,493 of the reports were also labeled by our reference annotator, LD — the same person whose labels we used for all the benchmarking. On those reports we can score each model against the human.

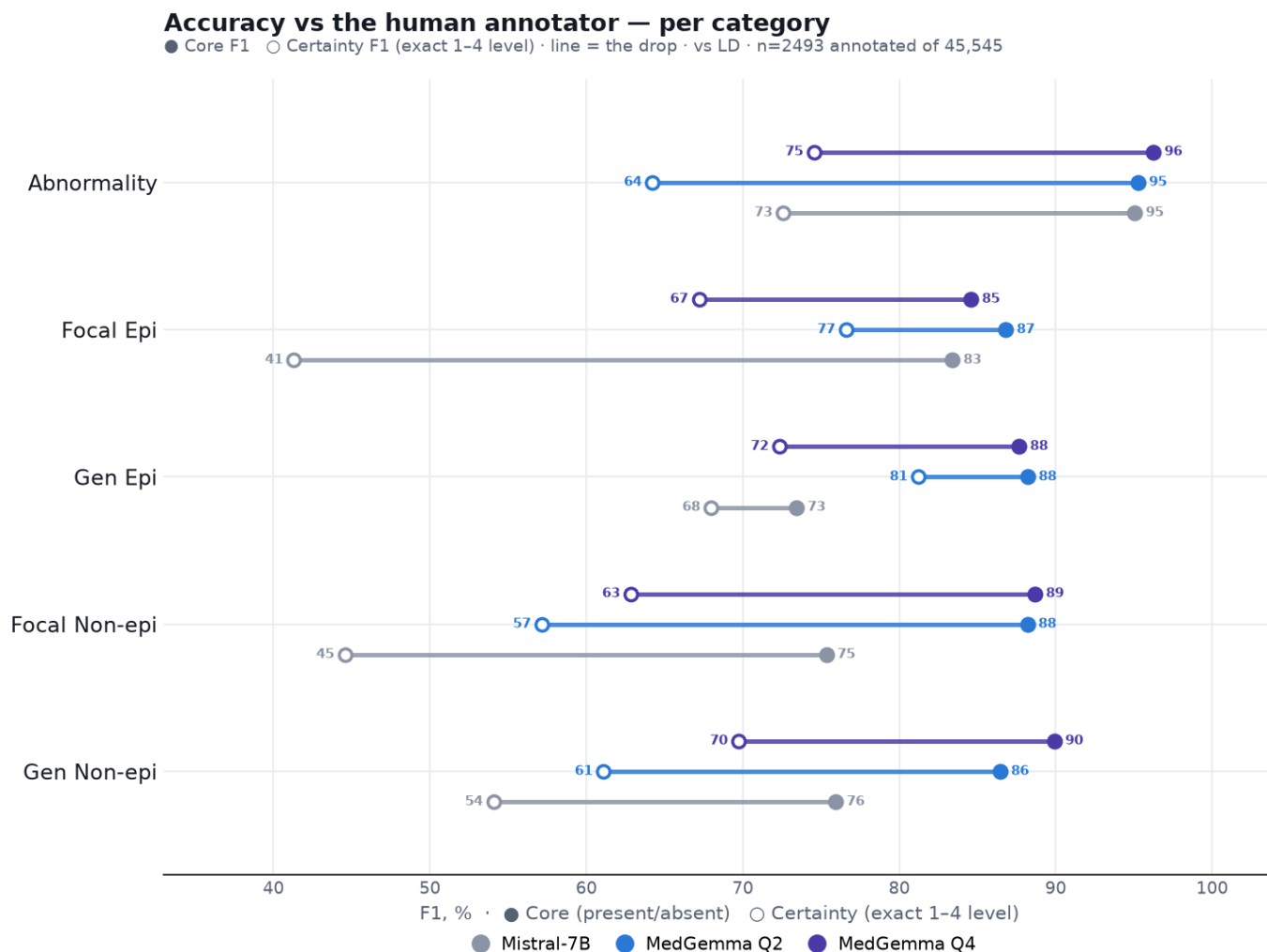


Figure 1: Accuracy vs the human annotator, per category

How to read the numbers: each value in the table is an F1 score for one finding — how well the model’s “present” calls line up with the human’s, balancing missed findings against false alarms, where 100 is

perfect. “Whole report” is the share of the 2,493 reports where the model got all five present/absent calls right.

Category	MedGemma Q4	MedGemma Q2	Mistral
Abnormality	96.2	95.3	95.1
Focal Epi	84.6	86.8	83.4
Gen Epi	87.7	88.2	73.4
Focal Non-epi	88.7	88.2	75.3
Gen Non-epi	89.9	86.4	75.9
Whole report	87.5%	87.1%	75.0%

Both models get the easy overall abnormal-or-normal call about right (~95). The gap opens up on the harder findings: for generalized epileptiform and the two slowing findings, our model is 13 to 14 points above Mistral; on focal epileptiform the two are about even. Those harder findings are also where the models disagree on the full set (next section), so the disagreements are mostly Mistral missing things.

In the chart, each model is a pair of dots per finding. The filled dot is the present/absent score above; the open dot is a stricter version that also requires the exact 1–4 confidence level to match, and the line is the drop between them. That stricter score is where Mistral falls apart — on focal epileptiform its confidence agreement drops to 41, while ours stays in the 70s.

Where the two models disagree (all 45,545)

Here there is no human label. We are just looking at how often each model calls each finding, and how often the two make the same call.

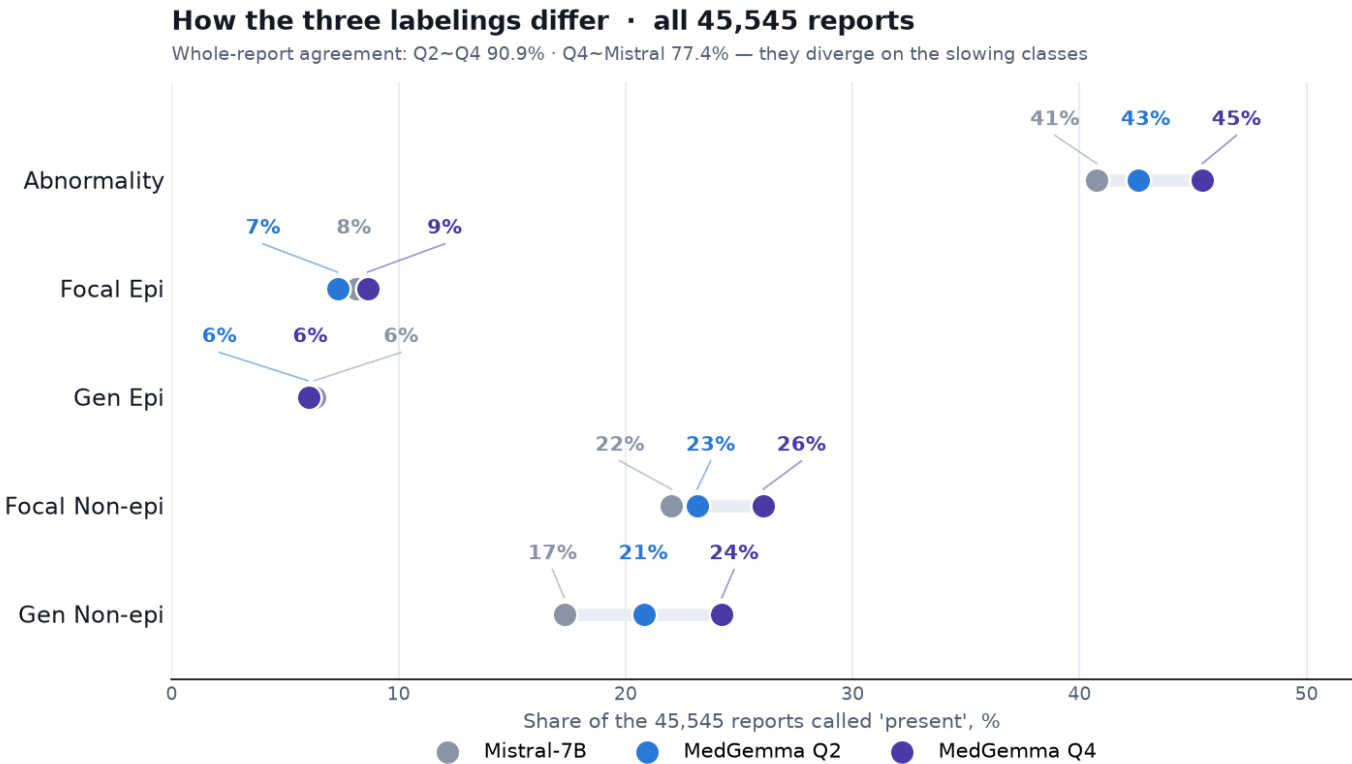


Figure 2: How the three labelings differ

How to read it: the “present” columns are the share of all 45,545 reports where that model scored the finding 3 or 4. “Agreement” is the share of reports where the two models made the same present/absent call for that finding. The whole-report figure is the share where all five calls match.

Category	MedGemma Q4 “present”	Mistral “present”	Agreement
Abnormality	45.4%	40.8%	93.3%
Focal Epi	8.6%	8.1%	98.0%
Gen Epi	6.0%	6.3%	98.3%
Focal Non-epi	26.1%	22.0%	88.9%
Gen Non-epi	24.2%	17.3%	89.6%

The two models make the same call on all five findings in **77.4%** of reports. They line up almost perfectly on the rare epileptiform findings and split on the slowing findings, where our model (especially Q4) flags more. Since that is also where our model is closer to the human, the extra flags are mostly real findings Mistral left out. The roughly one report in four where the two disagree is a good shortlist for a person to spot-check.

Does the model size matter? (Q2 vs Q4)

We labeled everything with the smaller Q2 model as well (~10 GB against ~15 GB). The two agree on 90.9% of reports, and where they split it is again the slowing findings.

Same reading as above: “present” columns are the share of the 45,545 each model scores 3 or 4; the last column is how often the two quantizations make the same present/absent call.

Category	Q2 “present”	Q4 “present”	Q2 vs Q4 agreement
Abnormality	42.6%	45.4%	96.6%
Focal Epi	7.3%	8.6%	98.6%
Gen Epi	6.0%	6.0%	99.1%
Focal Non-epi	23.2%	26.1%	96.1%
Gen Non-epi	20.8%	24.2%	95.8%

On the epileptiform findings the two are interchangeable (agreement above 98%). Q4 calls abnormality and slowing a little more often, and against the human that helps a bit — 87.5% whole-report against 87.1%, mostly from generalized slowing (89.9 against 86.4). That is the small gain on unfamiliar reports we expected from Q4, so Q4 is the labeling we are using and Q2 stays as a second opinion.

Pipeline, prompts, and the scripts behind this report: <https://github.com/vbronetskyi/eeg-report-classification>

How much to trust each label

Not every label is equally reliable, and the model gives us two independent ways to tell which ones to trust — neither of which needs a human to check.

The first is the model’s own confidence. Each finding is scored 1 to 4; a 1 or 4 is a confident call, a 2 or 3 is borderline. On the annotated reports, confident calls are right 99% of the time and borderline calls only 80%. Nearly all the errors sit in the borderline scores, which the model flags itself.

The second is agreement between the models. When Q4, Q2 and Mistral all make the same present/absent call, that call matches the human 96% of the time; when they split, only 65%. Across the whole set, 75% of reports are unanimous.

The two signals are independent, so putting them together sorts the labels cleanly:

The gold tier — confident *and* unanimous — is 99.4% correct and already covers 85% of all label decisions. At the level of whole reports, 60% of the 45k are gold (all five findings confident and unanimous) and come out 98.4% correct, while about 18% fall in the review tier at 55%. So most of the set can be used as it is, and a reviewer’s time goes to the ~18% that actually needs it.

The model knows when it is unsure

MedGemma Q4 · confident calls (1 & 4) are 99% right; borderline calls (2 & 3) only 80% · n=2,493 annotated

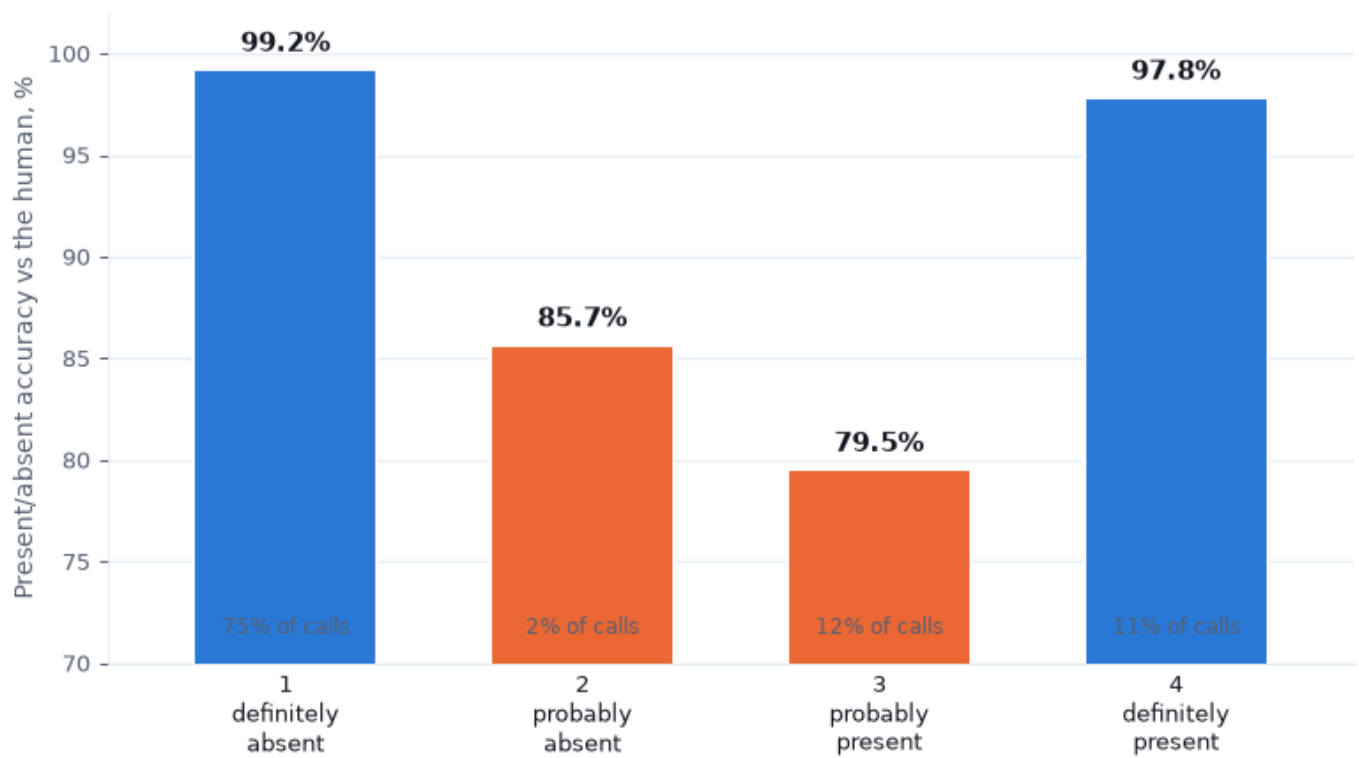


Figure 3: The model knows when it is unsure

Which labels to trust — two signals combined

Per-finding present/absent accuracy vs the human · top-right is the gold tier · MedGemma Q4 · n=2,493

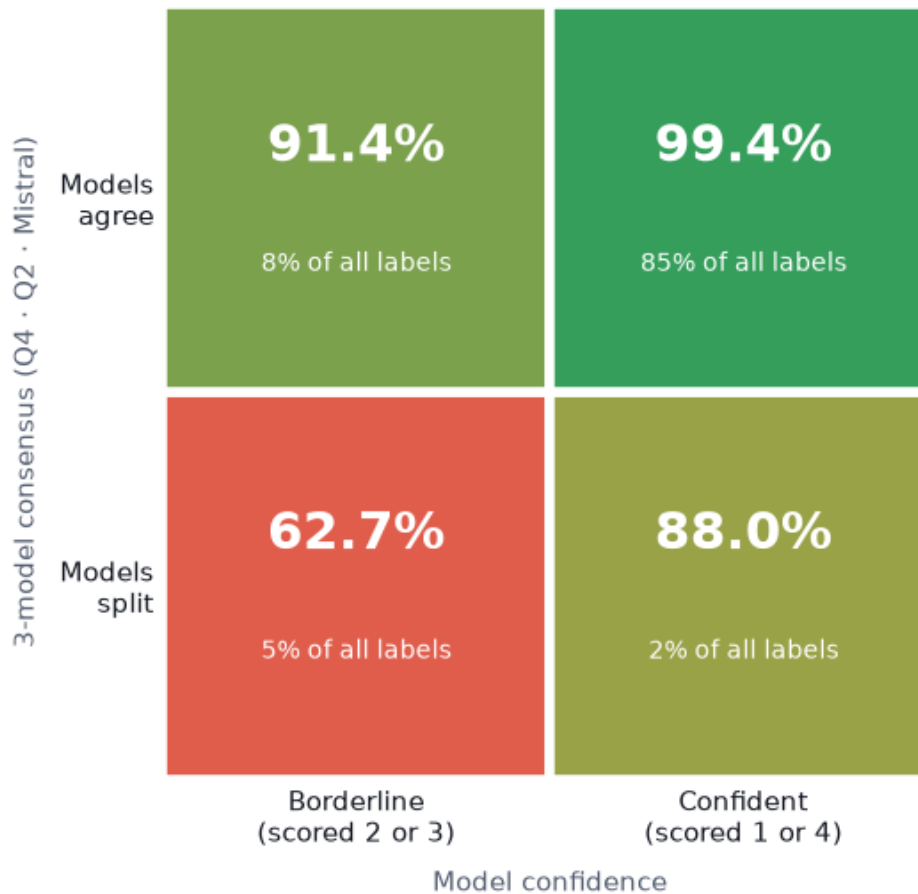


Figure 4: Which labels to trust